Stochastic Financial Models

Part II

Recorded from the lecturer in 2026-2027.

1 Martingales

We work with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

1.1 Definitions

A *filtration* is a sequence of σ-algebras $(\mathcal{F}_n)_{n\geq 0}$ such that

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1} \subseteq \mathcal{F}.$$

A *random process* is a sequence $X = (X_n)_{n\geq 0}$ of random variables.
Given a filtration $(\mathcal{F}_n)_{n\geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, we say that X is a *martingale* if:

(a) X is adapted, i.e. X_n is $\mathcal{F}_n$-measurable for all n.

(b) X is integrable, i.e. X_n is integrable for all n:

$$\mathbb{E}[|X_n|] < \infty.$$

(c) X satisfies the martingale property:

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \qquad \text{a.s., for all } n.$$

Given (a) and (b), condition (c) is equivalent to

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] = \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n.$$

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n \qquad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] \geq \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n\geq 0}$ is a *submartingale*.

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n \qquad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] \leq \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n\geq 0}$ is a *supermartingale*.

1.2 Natural filtration

Given a random process $X = (X_n)_{n\geq 0}$, its *natural filtration* $(\mathcal{F}_n^X)_{n\geq 0}$ is defined by

$$\begin{aligned}\mathcal{F}_n^X &= \sigma(X_0, X_1, \ldots, X_n)\\ &= \left\{\{(X_0, X_1, \ldots, X_n) \in B\} : B \in \mathcal{B}(\mathbb{R}^{n+1})\right\}.\end{aligned}$$

This is the smallest filtration making X adapted.

Exercise. Show that a martingale X with respect to $(\mathcal{F}_n)_{n\geq 0}$ is also a martingale with respect to $(\mathcal{F}_n^X)_{n\geq 0}$.

1.3 Examples

Let $X_1, X_2, \ldots$ be a sequence of independent integrable random variables. Set

$$\mathcal{F}_n = \mathcal{F}_n^X.$$

Define

$$S_0 = 0, \qquad Z_0 = 1,$$

and

$$S_n = X_1 + \cdots + X_n, \qquad Z_n = \prod_{i=1}^{n} X_i.$$

- If $\mathbb{E}[X_n] = 0$ for all n, then $(S_n)_{n\geq 0}$ is a martingale.
- If $X_n \geq 0$ and $\mathbb{E}[X_n] = 1$ for all n, then $(Z_n)_{n\geq 0}$ is a martingale.

  To see this, take $A \in \mathcal{F}_n$. Then

$$\begin{aligned}\mathbb{E}[S_{n+1}\mathbf{1}_A] &= \mathbb{E}[X_{n+1}\mathbf{1}_A] + \mathbb{E}[S_n\mathbf{1}_A]\\ &= \underbrace{\mathbb{E}[X_{n+1}]\mathbb{P}(A)}_{=0} + \mathbb{E}[S_n\mathbf{1}_A]\\ &= \mathbb{E}[S_n\mathbf{1}_A].\end{aligned}$$

  Also,

$$\begin{aligned}\mathbb{E}[Z_{n+1}\mathbf{1}_A] &= \mathbb{E}[X_{n+1}Z_n\mathbf{1}_A]\\ &= \underbrace{\mathbb{E}[X_{n+1}]}_{=1}\mathbb{E}[Z_n\mathbf{1}_A]\\ &= \mathbb{E}[Z_n\mathbf{1}_A].\end{aligned}$$

This also shows integrability.

2 Conditional Probabilities and Expectation

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

and

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X\mathbf{1}_B]}{\mathbb{P}(B)}.$$

2.1 Existence and uniqueness

Theorem. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra. Let X be an integrable random variable. Then there exists a random variable Y such that:

(a) Y is $\mathcal{G}$-measurable;

(b)
$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A] \qquad \text{for all } A \in \mathcal{G}.$$

Moreover, if Y' is another random variable satisfying (a) and (b), then

$$Y = Y' \qquad \text{a.s.}$$

Proof of uniqueness. Let X' be another integrable random variable with $X \leq X'$. Suppose Y' satisfies (a) and (b) with respect to X'.

Set
$$A = \{Y > Y'\}, \qquad Z = (Y - Y')\mathbf{1}_A.$$

Then $A \in \mathcal{G}$ and $Z \geq 0$. Hence

$$\begin{aligned}
\mathbb{E}[Z] &= \mathbb{E}[(Y - Y')\mathbf{1}_A] \\
&= \mathbb{E}[Y\mathbf{1}_A] - \mathbb{E}[Y'\mathbf{1}_A] \\
&= \mathbb{E}[X\mathbf{1}_A] - \mathbb{E}[X'\mathbf{1}_A] \\
&= \mathbb{E}[(X - X')\mathbf{1}_A] \leq 0.
\end{aligned}$$

Therefore $Z = 0$ a.s., so
$$Y \leq Y' \qquad \text{a.s.}$$

Taking $X' = X$, we also obtain $Y \geq Y'$ a.s. Thus

$$Y = Y' \qquad \text{a.s.}$$

2.2 Notation

We write
$$Y = \mathbb{E}[X \mid \mathcal{G}] \qquad \text{a.s.}$$

If $\mathcal{G} = \sigma(W)$, where W is a random variable, we write

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X \mid W].$$

If $X = \mathbf{1}_A$, we write
$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{P}(A \mid \mathcal{G}).$$

End of Lecture 1

Optional stopping

We say that a random time

$$T : \Omega \longrightarrow \{0, 1, 2, \ldots\} \cup \{\infty\}$$

is a **stopping time** if

$$\{T \leq n\} \in \mathcal{F}_n \qquad \text{for all } n.$$

Example. Let $(X_n)_{n\geq 0}$ be an adapted process, with values in $\mathbb{Z}$, say. Set

$$T = \inf\{n \geq 0 : X_n = 0\}.$$

Then

$$\{T \leq n\} = \underbrace{\{X_0 = 0\}}_{\in \mathcal{F}_0} \cup \underbrace{\{X_1 = 0\}}_{\in \mathcal{F}_1} \cup \cdots \cup \underbrace{\{X_n = 0\}}_{\in \mathcal{F}_n},$$

with

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_n.$$

So T is a stopping time.

Optional stopping theorem

Let $(M_n)_{n\geq 0}$ be a martingale and let T be a bounded stopping time. Then

$$\mathbb{E}(M_T) = \mathbb{E}(M_0), \qquad \big(M_T(\omega) = M_{T(\omega)}(\omega)\big).$$

Proof. Choose n such that $T \leq n$. Then

$$\begin{aligned} M_T &= M_0 + (M_1 - M_0) + \cdots + (M_T - M_{T-1}) \\ &= M_0 + \sum_{k=1}^{n} (M_k - M_{k-1}) \mathbf{1}_{\{T \geq k\}}. \end{aligned}$$

Now

$$\{T \geq k\} = \{T \leq k-1\}^c \in \mathcal{F}_{k-1},$$

so

$$\mathbb{E}\big((M_k - M_{k-1})\mathbf{1}_{\{T\geq k\}}\big) = 0 \qquad \text{for all } k.$$

Take expectations to see

$$\mathbb{E}(M_T) = \mathbb{E}(M_0) + 0. \qquad \square$$

OST II

Let $(M_n)_{n\geq 0}$ be a martingale and let T be a finite-valued stopping time. Suppose there exists $C < \infty$ such that one of the following holds:

(a)

$$|M_n| \leq C \qquad \text{for all } n \leq T.$$

(b)

$$\mathbb{E}(T) < \infty \qquad \text{and} \qquad |M_n - M_{n-1}| \leq C \qquad \text{for all } n \leq T.$$

Then

$$\mathbb{E}(M_T) = \mathbb{E}(M_0).$$

Proof. Define

$$T_n = T \wedge n.$$

Then T_n is a bounded stopping time. Since $T < \infty$, $T_n = T$ eventually, so

$$M_{T_n} \longrightarrow M_T \qquad \text{as } n \longrightarrow \infty.$$

By optional stopping, we have

$$\mathbb{E}(M_{T_n}) = \mathbb{E}(M_0).$$

If (a) holds, then

$$\sup_n |M_{T_n}| \leq C,$$

so

$$\mathbb{E}(M_{T_n}) \longrightarrow \mathbb{E}(M_T)$$

by bounded convergence.

If (b) holds, then

$$\sup_n |M_{T_n}| \leq CT + |M_0| \qquad \text{and} \qquad \mathbb{E}(T) < \infty,$$

so

$$\mathbb{E}(M_{T_n}) \longrightarrow \mathbb{E}(M_T)$$

by dominated convergence.

In any case, we can let $n \longrightarrow \infty$ to see

$$\mathbb{E}(M_T) = \mathbb{E}(M_0). \qquad \square$$

Examples

Let $X_1, X_2, \ldots$ and $Y_1, Y_2, \ldots$ be independent random variables such that

$$\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2}, \qquad \mathbb{P}(Y_n = 0) = \mathbb{P}(Y_n = 2) = \frac{1}{2}.$$

Set

$$S_n = 1 + X_1 + \cdots + X_n, \qquad Z_n = \prod_{i=1}^{n} Y_i, \qquad (Z_0 = 1).$$

Then $(S_n)_{n\geq 0}$ and $(Z_n)_{n\geq 0}$ are martingales.

Set

$$T = \inf\{n \geq 0 : S_n = 0\}.$$

Then

$$\mathbb{P}(T < \infty) = 1,$$

but

$$\mathbb{E}(S_0) = 1, \qquad \mathbb{E}(S_T) = 0.$$

Now set

$$T = \inf\{n \geq 0 : Z_n = 0\}.$$

Then

$$\mathbb{P}(T \leq n) = 1 - 2^{-n},$$

so

$$\mathbb{P}(T < \infty) = 1.$$

In fact,

$$\mathbb{E}(T) < \infty.$$

But

$$\mathbb{E}(Z_T) = 0, \qquad \mathbb{E}(Z_0) = 1.$$

Recall that, for a σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ and an integrable random variable X, $\mathbb{E}(X \mid \mathcal{G})$ is characterized by

$$Y = \mathbb{E}(X \mid \mathcal{G}) \qquad \text{a.s.}$$

if

(a) Y is $\mathcal{G}$-measurable;

(b) Y is integrable and

$$\mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(Y\mathbf{1}_A) \qquad \text{for all } A \in \mathcal{G}.$$

Properties of conditional expectation

(i)

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{G})) = \mathbb{E}(X).$$

(ii) If X is $\mathcal{G}$-measurable, then

$$\mathbb{E}(X \mid \mathcal{G}) = X \qquad \text{a.s.}$$

(iii) If X is independent of $\mathcal{G}$, then

$$\mathbb{E}(X \mid \mathcal{G}) = \mathbb{E}(X) \qquad \text{a.s.}$$

(iv) If $X \geq 0$ a.s., then

$$\mathbb{E}(X \mid \mathcal{G}) \geq 0 \qquad \text{a.s.}$$

(v) **Linearity.** For W, X integrable and $\alpha, \beta \in \mathbb{R}$,

$$\mathbb{E}(\alpha X + \beta W \mid \mathcal{G}) = \alpha\mathbb{E}(X \mid \mathcal{G}) + \beta\mathbb{E}(W \mid \mathcal{G}).$$

Take

$$Y = \mathbb{E}(X \mid \mathcal{G}), \qquad U = \mathbb{E}(W \mid \mathcal{G}).$$

Then $\alpha Y + \beta U$ is $\mathcal{G}$-measurable and integrable.

For $A \in \mathcal{G}$,

$$\begin{aligned}\mathbb{E}((\alpha X + \beta W)\mathbf{1}_A) &= \alpha\mathbb{E}(X\mathbf{1}_A) + \beta\mathbb{E}(W\mathbf{1}_A) \\ &= \alpha\mathbb{E}(Y\mathbf{1}_A) + \beta\mathbb{E}(U\mathbf{1}_A) \\ &= \mathbb{E}((\alpha Y + \beta U)\mathbf{1}_A)\,.\end{aligned}$$

Hence

$$\alpha Y + \beta U = \mathbb{E}(\alpha X + \beta W \mid \mathcal{G}) \qquad \text{a.s.}$$

Proposition (Tower property)

Let $\mathcal{H}$ be another σ-algebra, with

$$\mathcal{H} \subseteq \mathcal{G}.$$

Then

$$\mathbb{E}(\mathbb{E}(X \mid \mathcal{G}) \mid \mathcal{H}) = \mathbb{E}(X \mid \mathcal{H}).$$

Proof. Take

$$Y = \mathbb{E}(\mathbb{E}(X \mid \mathcal{G}) \mid \mathcal{H}) \qquad \text{a.s.}$$

Then, for $A \in \mathcal{H}$, we have $A \in \mathcal{G}$, so

$$\mathbb{E}(Y\mathbf{1}_A) \underset{A\in\mathcal{H}}{=} \mathbb{E}(\mathbb{E}(X \mid \mathcal{G})\mathbf{1}_A) \underset{A\in\mathcal{G}}{=} \mathbb{E}(X\mathbf{1}_A).$$

Y is integrable and $\mathcal{H}$-measurable, so

$$Y = \mathbb{E}(X \mid \mathcal{H}) \qquad \text{a.s.} \qquad \square$$

End of Lecture 2

Properties of conditional expectation (continued)

Proposition (Taking out what is known). Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra, and let X be an integrable random variable. Suppose that Z is a bounded $\mathcal{G}$-measurable random variable. Then

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}] \qquad \text{a.s.}$$

Proof. Take $Y = \mathbb{E}[X \mid \mathcal{G}]$ a.s. Note that ZY is integrable and $\mathcal{G}$-measurable.

Take $A \in \mathcal{G}$. We will show that

$$\mathbb{E}[ZX\mathbf{1}_A] = \mathbb{E}[ZY\mathbf{1}_A],$$

so that ZY satisfies (a) and (b) for ZX, and we are done.

Consider first the case $Z = \mathbf{1}_B$, where $B \in \mathcal{G}$. Then $A \cap B \in \mathcal{G}$, so

$$\begin{aligned}\mathbb{E}[ZX\mathbf{1}_A] &= \mathbb{E}[X\mathbf{1}_{A\cap B}] \\ &= \mathbb{E}[Y\mathbf{1}_{A\cap B}] \\ &= \mathbb{E}[ZY\mathbf{1}_A].\end{aligned}$$

This extends to simple random variables

$$Z = \sum_{i=1}^{n} a_i \mathbf{1}_{B_i}$$

by linearity.

For $Z \geq 0$ and $X \geq 0$, consider

$$Z_n = \left(2^{-n} \lfloor 2^n Z \rfloor\right) \wedge n.$$

Then Z_n is simple, so

$$\mathbb{E}[Z_n X\mathbf{1}_A] = \mathbb{E}[Z_n Y\mathbf{1}_A].$$

Since $X \geq 0$, we have $Y \geq 0$ a.s., so

$$Z_n X \uparrow ZX, \qquad Z_n Y \uparrow ZY.$$

Letting $n \to \infty$,

$$\mathbb{E}[Z_n X \mathbf{1}_A] \longrightarrow \mathbb{E}[ZX\mathbf{1}_A],$$
$$\mathbb{E}[Z_n Y \mathbf{1}_A] \longrightarrow \mathbb{E}[ZY\mathbf{1}_A].$$

Finally, for general X and Z, use

$$X = X^+ - X^-, \qquad Z = Z^+ - Z^-,$$

and linearity to obtain the desired formula. □

Proposition. Let X_1, X_2 be random variables, with X_1 being $\mathcal{G}$-measurable and X_2 independent of $\mathcal{G}$. Suppose that F is a bounded measurable function on $\mathbb{R}^2$. Then

$$\mathbb{E}[F(X_1, X_2) \mid \mathcal{G}] = f(X_1) \qquad \text{a.s.},$$

where

$$f(x) = \mathbb{E}[F(x, X_2)].$$

Recall the optional stopping theorem. Let M be a martingale and let T be a bounded stopping time. Then

$$\mathbb{E}[M_T] = \mathbb{E}[M_0].$$

Theorem. Let M be a martingale and let T be a stopping time. Consider the stopped process

$$M_n^T := M_{T \wedge n}.$$

Then M^T is also a martingale.

Proof. We have

$$M_n^T = M_0 + \sum_{k=1}^{n} (M_k - M_{k-1}) \mathbf{1}_{\{k \le T\}}.$$

Since $\{k \le T\} \in \mathcal{F}_k$, this shows that M_n^T is $\mathcal{F}_n$-measurable. Also, M^T is integrable.

To show the martingale property, it suffices to show, for $A \in \mathcal{F}_n$ and all n, that

$$\mathbb{E}\left[(M_{n+1}^T - M_n^T)\mathbf{1}_A\right] = 0.$$

But

$$M_{n+1}^T - M_n^T = (M_{n+1} - M_n)\mathbf{1}_{\{n+1 \le T\}}.$$

Note that

$$\{n+1 \le T\} = \{T \le n\}^c \in \mathcal{F}_n.$$

So

$$\mathbb{E}\left[(M_{n+1} - M_n) \underbrace{\mathbf{1}_{\{n+1 \le T\}} \mathbf{1}_A}_{\mathcal{F}_n\text{-measurable}}\right] = 0,$$

as required. □

Martingale transforms

Say that a process $(H_n)_{n\geq 1}$ is *previsible* if H_n is $\mathcal{F}_{n-1}$-measurable for all n.

Theorem. Let M be a martingale and let H be a bounded previsible process. We define the transform $H \cdot M$ of M by H by

$$(H \cdot M)_n = Y_n = \sum_{k=1}^{n} H_k(M_k - M_{k-1}), \qquad n \geq 1,$$

and

$$(H \cdot M)_0 = Y_0 = 0.$$

Interpretation. Think of M_n as the price of an asset at time n.
An investor holds H_n units of the asset between time $n-1$ and time n.
Then the profit made from time 0 to time n is Y_n.
The previsible restriction is natural.

Then $H \cdot M$ is also a martingale. In particular,

$$\mathbb{E}[(H \cdot M)_n] = \mathbb{E}[(H \cdot M)_0] = 0.$$

Proof. Note that

$$(H \cdot M)_{n+1} - (H \cdot M)_n = H_{n+1}(M_{n+1} - M_n),$$

and, by taking out what is known,

$$\begin{aligned} &\mathbb{E}\left[H_{n+1}(M_{n+1} - M_n) \mid \mathcal{F}_n\right] \\ &\quad = H_{n+1} \underbrace{\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n]}_{=0, \text{ as } M \text{ is a martingale}}. \end{aligned}$$

So $H \cdot M$ is a martingale, as claimed. □

End of Lecture 3